WORKED SOLUTION

STEP I 1989, Question 6

Pure • Normals and Differential Equations

STEP I 1989, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q6

The key idea

The normal at P has gradient $-1/\mathrm{f}'(x)$, so finding Q is just a matter of putting $X = 0$ into its equation. Then PQ^2 is a horizontal distance squared plus a vertical one, and the given length turns into an equation for $\mathrm{f}'(x)$ alone. The only subtlety is the square root at the end: the sign we choose is decided by the minimum point.

Solution

The y-coordinate of Q

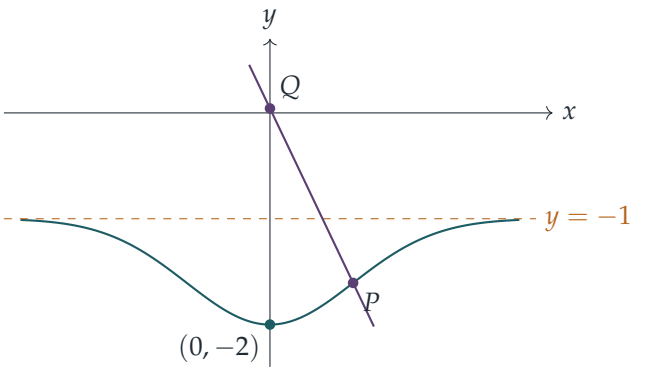
At $P = (x, \mathrm{f}(x))$ the tangent has gradient $\mathrm{f}'(x)$, so (provided $\mathrm{f}'(x) \neq 0$) the normal has gradient $-1/\mathrm{f}'(x)$. Using (X, Y) for the running coordinates, the normal is

$$Y - \mathrm{f}(x) = -\frac{1}{\mathrm{f}'(x)}\,(X - x).$$

Putting $X = 0$, the normal meets the y-axis at

$$Y = \mathrm{f}(x) + \frac{x}{\mathrm{f}'(x)}, \qquad \text{so} \qquad Q = \left(0,\ \mathrm{f}(x) + \frac{x}{\mathrm{f}'(x)}\right).$$

(If $\mathrm{f}'(x) = 0$ the normal is vertical and, unless $x = 0$, it never meets the y-axis.)



The differential equation

P and Q differ by x horizontally and by $x/\mathrm{f}'(x)$ vertically, so

$$PQ^2 = x^2 + \frac{x^2}{(\mathrm{f}'(x))^2}.$$

Setting this equal to $\mathrm{e}^{x^2} + x^2$, the x^2 terms cancel and we get

$$\frac{x^2}{(\mathrm{f}'(x))^2} = \mathrm{e}^{x^2}, \qquad \text{that is} \qquad \boxed{(\mathrm{f}'(x))^2 = x^2\,\mathrm{e}^{-x^2}.}$$

Equivalently $\mathrm{f}'(x) = \pm x\,\mathrm{e}^{-x^2/2}$.

The curve with a minimum at $(0, -2)$

Ask yourself Which sign do we take, and could the sign change part-way along the curve?

Since f' is continuous (the curve has a minimum point, so we are assuming it is differentiable) and $\mathrm{f}'(x)^2 = x^2\mathrm{e}^{-x^2}$ vanishes only at $x = 0$, f' cannot change sign on $x > 0$ or on $x < 0$: it would have to pass through zero. For $(0, -2)$ to be a minimum, f must be decreasing just to the left of 0 and increasing just to the right, so $\mathrm{f}'(x) < 0$ for $x < 0$ and $\mathrm{f}'(x) > 0$ for $x > 0$. That is exactly the sign of x, so on both sides

$$\mathrm{f}'(x) = x\,\mathrm{e}^{-x^2/2}.$$

(The minus sign throughout would give a maximum at $x = 0$; mixing the signs would give f' of constant sign, i.e. a point of inflection.)

Integrating, since $\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-x^2/2}) = -x\,\mathrm{e}^{-x^2/2}$,

$$\mathrm{f}(x) = -\mathrm{e}^{-x^2/2} + c.$$

The condition $\mathrm{f}(0) = -2$ gives $-1 + c = -2$, so $c = -1$ and the curve is

$$\boxed{y = -1 - \mathrm{e}^{-x^2/2}.}$$

It has its minimum value -2 at $x = 0$ and rises towards the asymptote $y = -1$ on both sides, as in the sketch.

Checks

- At $x = 1$: $\mathrm{f}(1) = -1 - \mathrm{e}^{-1/2} \approx -1.6065$ and $\mathrm{f}'(1) = \mathrm{e}^{-1/2} \approx 0.6065$, so $Q \approx (0,\, 0.0422)$ and $PQ \approx \sqrt{1 + 1.6487^2} \approx 1.9283$, which equals $\sqrt{\mathrm{e} + 1}$, as required.
- The computer recomputed Q by intersecting the normal line with the y-axis at 2000 random points and found PQ agreeing with $\sqrt{\mathrm{e}^{x^2} + x^2}$ to within 10^{-15} (relative).

Common slips

- Using the tangent's gradient $\mathrm{f}'(x)$ instead of the normal's $-1/\mathrm{f}'(x)$, which gives Q at $\mathrm{f}(x) - x\mathrm{f}'(x)$.
- Taking the square root as $\mathrm{f}'(x) = x\mathrm{e}^{-x^2/2}$ without saying why the minus sign is rejected.

Beyond STEP

Here a geometric condition on every point of a curve became a first-order differential equation, but one in which f' appears squared, so it does not determine a unique solution until extra information picks the branch. Such equations, not solved for y', are where uniqueness theorems for ODEs start to need care, and they lead on to envelopes and singular solutions. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and their existence and uniqueness theory.

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